

Data Analytics Report:

La Liga Primera División 2025–2026

CMPT 391 – Database Management Systems

Section AS01 | Project II

MacEwan University

Winter 2026

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1. Introduction

This report applies two complementary analytical methods to La Liga match data from previous seasons. First, a supervised classification analysis was conducted to predict full-time match outcomes using pre-match features derived from historical team performance and implied probabilities. Second, an unsupervised clustering analysis was performed to assess whether teams could be meaningfully partitioned into distinct performance profiles based on their aggregate statistical characteristics across the season.

2. Data Wrangling

The source dataset was obtained from Football-Data.co.uk [2] and contained match-level records for La Liga Primera División spanning multiple seasons (2018–2026) and processed in Python. In its raw form, the dataset comprised 2,950 rows and 148 columns, the majority of which were not required for the project objectives.

2.1 Column Selection

As a first step, 124 bookmaker-related columns were removed, retaining only the Bet365 closing odds alongside the core match-level attributes: team names, date, full-time and half-time results and goals, shots, shots on target, corners, fouls, yellow cards, and red cards. This resulted in reducing the working dataset to 31 columns.

2.2 Integrity Checks and Type Enforcement

Date values were parsed to handle the day-first format used by Football-Data.co.uk. Rows with missing values in any key field such as date, team names, or full-time result, were dropped, as such records cannot be assigned to a match. For the remaining numeric statistics, missing values were imputed using each column’s median. The dataset was then sorted chronologically by date in preparation for the rolling-window feature engineering described in section 3.1.

2.3 Derived Difference Features

Seven within-match difference variables were computed to summarize the play between the home and away side: *shot difference*, *shots-on-target difference*, *corner difference*, *foul difference*, *yellow-card difference*, and *red-card difference* (each defined as the home value minus

the away value). These features are retained in the cleaned match-level dataset and serve as supplementary descriptors in both the classification and clustering pipelines.

2.4 Downstream Output

The preparation stage produced two structured datasets from the same cleaned source. For the classification analysis (Section 3), the match-level dataset (2,950 rows) served as the basis for rolling-window feature engineering, which reduced the usable sample to matches where each team had at least five prior games; the full feature engineering procedure is described in Section 3.1.

For the clustering analysis (Section 4), records were aggregated from the match level to the team level so that each row represented a single La Liga club yielding a 28-row dataset. Team-level averages were computed for goals scored, goals conceded, shots, shots on target, corners, fouls, yellow cards, and red cards, together with home and overall win rates. The identifier variable *team* was excluded from the model inputs, and *total_games* was omitted as it is approximately constant across clubs and provides no discriminating signal. All remaining features were standardized prior to K-means clustering.

3. Match Outcome Classification

3.1 Research Design and Methodology

The objective of this analysis was to build a supervised classification model capable of predicting the result (FTR) of a La Liga match as a Home win (H), Draw (D), or Away win (A), using only information available before kickoff.

For each game, a rolling window of the last five matches was computed per team to capture recent form, including average goals scored and conceded, shots, shots on target, corners, fouls, and win rate along with rest days and a head-to-head historical win rate for each specific matchup. Separate home and away win rates were also computed, as teams in La Liga exhibit substantially different performance profiles at home versus away (Figure 1).

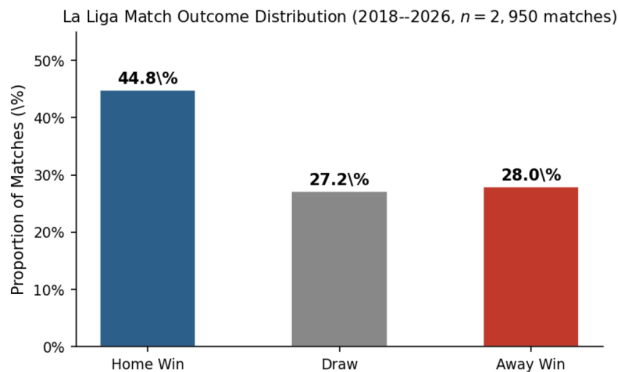


Figure 1: Home vs. Away Distribution

The final dataset counted 2,859 rows after excluding early-season matches where teams had fewer than five prior games. The dataset was split chronologically: the earliest 80% of matches (2,287) formed the training set, and the most recent 20% (572 matches, primarily from the 2025/26 season) served as the test set.

From there, five classification algorithms were tested: Logistic Regression, Naive Bayes, Decision Tree (max depth = 5), Random Forest (100 estimators), and an IF-THEN Rules classifier. To address the imbalance between Home wins, Draws, and Away wins, each had `class_weight='balanced'` where applicable Naive Bayes estimates the posterior probability of each outcome class using Bayes' theorem, assuming conditional independence between features given the class label, with each feature modelled as a Gaussian distribution. The IF-THEN Rules classifier implements explicit conditional rules based on the implied win probabilities, the strongest predictive signal in the feature set. Thresholds for the home and away implied probabilities were fitted by grid search on the training data, producing rules of the form: if implied home probability exceeds a threshold, predict a Home win; else if implied away probability exceeds a threshold, predict an Away win; otherwise predict a Draw.

3.2 Results

A naïve baseline of always predicting the Home win achieved 46.0% accuracy on the test set, consistent with the well-documented home advantage effect in football [5].

Table 1: Classification model test accuracy compared to the naïve baseline

Model	Test Accuracy
Naïve Baseline (always Home win)	46.0%
Logistic Regression	51.9%
Naive Bayes	50.2%
Decision Tree	47.2%
Random Forest	53.7%
IF-THEN Rules	54.0%

The IF-THEN Rules classifier achieved the highest accuracy at 54.0%, exceeding the baseline by 8.0 percentage points. Despite its simplicity, the rules-based approach outperformed all other models by leveraging the implied probabilities directly.

The Naive Bayes classifier achieved 50.2% accuracy, competitive with Logistic Regression despite its conditional independence assumption, consistent with prior observations that Naive Bayes performs comparably to more complex classifiers across many real-world domains [4].

The Random Forest feature importance revealed that the Bet365 implied probabilities were the strongest predictors, followed by the home team’s rolling goal difference and win rate. Fouls and corners contributed minimally, suggesting limited predictive value (Figure 2).

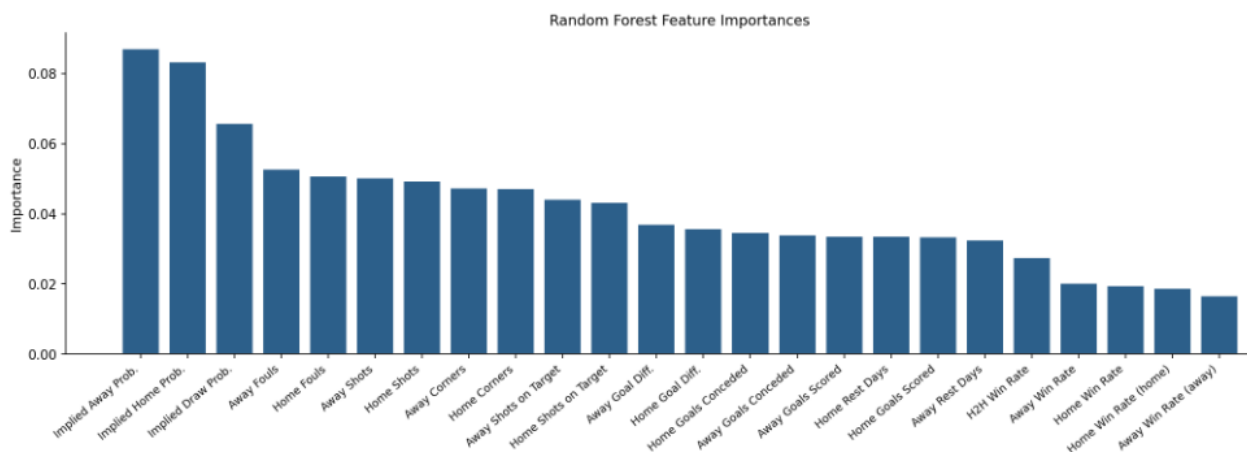


Figure 2: Feature Importance

Draws proved consistently the most difficult class to predict across all five models (Figure 3). This is consistent with the broader literature; draws represent a high-entropy outcome that depends on in-match dynamics difficult to capture from pre-match features alone [1].

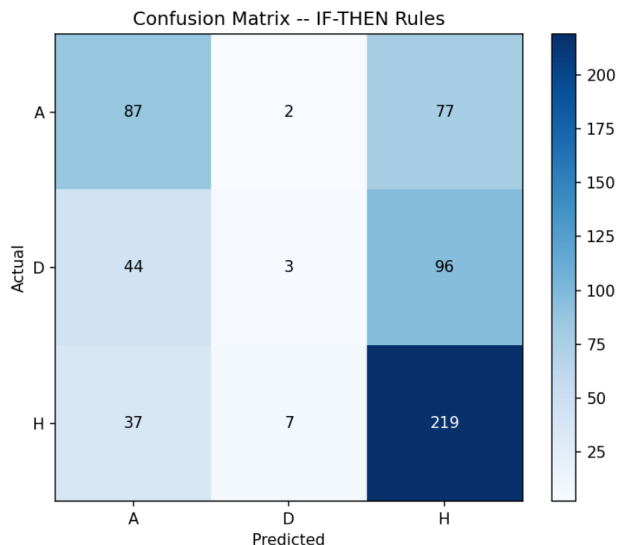


Figure 3: Confusion Matrix

4. Team Performance Clustering

4.1 Design and Methodology

The purpose of this analysis was to examine whether La Liga teams could be grouped into distinct profiles based on their average statistical characteristics. Unlike the classification analysis, which focused on predicting match outcomes, this part of the study uses an unsupervised learning approach to identify natural groupings among teams without predefined labels.

A K-Means clustering analysis was conducted using standardized team-level variables related to attacking output, defensive outcomes, discipline, and success rates. These variables were chosen to capture multiple dimensions of team performance rather than relying only on league position or points.

An elbow plot was inspected to assess the number of clusters, and three clusters were selected because they provided a reasonable balance between separation and interpretability

(Figure 4).

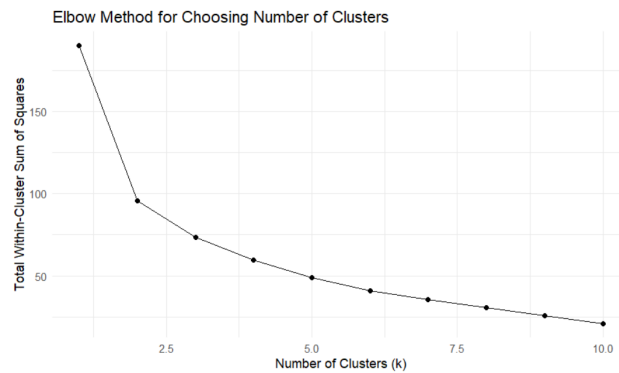


Figure 4: Elbow Plot

To support interpretation of the results, Principal Component Analysis (PCA) was used to project the teams into two dimensions for visualization (Figure 5).

4.2 Results

The clustering results suggest that the teams can be divided into three broad groups, as summarized in Table 2.

Table 2: Clustering summary: average statistics by cluster

Variable	Cluster 1	Cluster 2	Cluster 3
Avg. goals scored	2.18	0.97	1.30
Avg. goals conceded	0.94	1.41	1.43
Avg. shots	17.1	11.2	12.0
Avg. shots on target	6.54	3.66	4.13
Avg. corners	6.77	4.77	4.32
Avg. fouls	10.1	14.3	12.4
Avg. yellow cards	1.72	2.61	2.14
Avg. red cards	0.13	0.21	0.12
Home win rate	0.91	0.30	0.45

Cluster 1 showed the strongest overall performance, with the highest average goals scored

(2.18), the lowest average goals conceded (0.94), the highest number of shots (17.1), and the highest home win rate (0.91). This cluster represents the strongest attacking and highest-performing teams.

Cluster 2 showed weaker offensive output and lower success rates, with fewer goals scored (0.97), fewer shots on target (3.66), and a lower home win rate (0.30). This group also had the highest average fouls (14.3) and yellow cards (2.61), suggesting a more physical or less disciplined style of play.

Cluster 3 appeared to represent mid-level teams, with moderate values across most variables, including goals scored (1.30), shots (12.0), and home win rate (0.45). These teams were not as dominant as those in Cluster 1, but they also did not perform as poorly as those in Cluster 2.

The PCA visualization (Figure 5) supports the clustering results by showing a reasonable separation among the three groups. In particular, Cluster 1 contains Real Madrid, Barcelona, and Atlético Madrid, which are clearly separated from most of the other teams. This is consistent with their stronger attacking statistics and higher win rates. The remaining teams are distributed between Clusters 2 and 3, reflecting weaker or more balanced performance profiles.

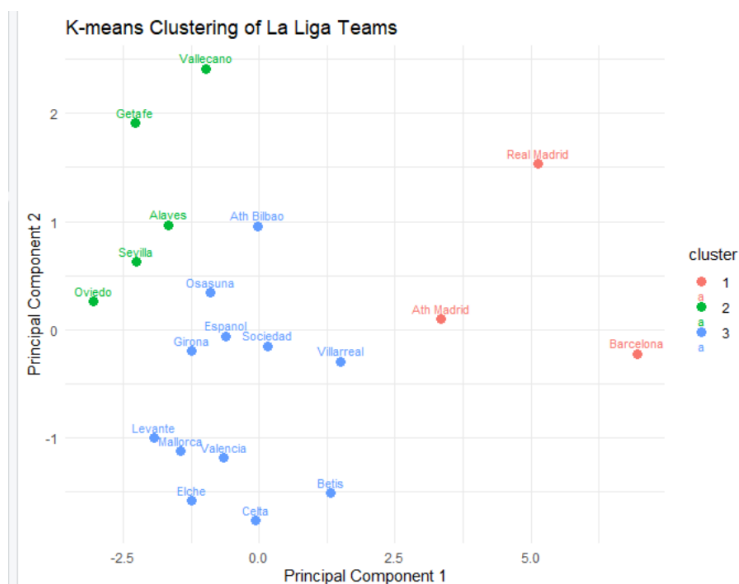


Figure 5: PCA Visualization

Overall, the clustering analysis suggests that La Liga teams can be meaningfully grouped based on their statistical performance profiles in a manner beyond simple league rankings.

5. Conclusion

This report demonstrated that La Liga data can support both predictive and descriptive analysis. The classification analysis showed that match outcomes can be predicted above the naïve home-win baseline, with the IF-THEN Rules classifier achieving 54.0% accuracy by leveraging implied probabilities. The clustering analysis grouped the 28 teams into three distinct profiles; dominant attacking teams, weaker physical teams, and balanced mid-level sides, confirming that meaningful performance tiers exist within the league beyond simple standings.

Both analyses carry limitations. The classification model covers a single league and may not generalize to competitions with different structures; the 2025/26 test set is also incomplete (290 of 380 matches played), so it does not represent a full season’s distribution. Key contextual factors including injuries, lineup changes, and managerial decisions are absent from the dataset, limiting accuracy to what can be inferred from statistics and odds alone [3]. A potential sampling bias also exists: squad compositions shift between seasons, and a model trained on 2018–2025 data may not fully reflect 2025/26 conditions. For clustering, team-level averages mask match-to-match variation, and the choice of three clusters and the selected variables directly shape the resulting groups; the profiles should therefore be treated as broad, season-specific summaries rather than fixed categories.

5. References

- [1] Dixon, M. J., & Coles, S. G. (1997). Modelling association football scores and inefficiencies in the football betting market. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 46(2), 265–280.
- [2] Football-Data.co.uk. (2026). *Spain: La Liga Primera Division (season 2025/2026) results, match statistics, and betting odds data*. Retrieved from <https://www.football-data.co.uk/spainm.php>
- [3] Goddard, J. (2005). Regression models for forecasting goals and match results in association football. *International Journal of Forecasting*, 21(2), 331–340.
- [4] Hand, D. J., & Yu, K. (2001). Idiot’s Bayes — not so stupid after all? *International Statistical Review*, 69(3), 385–398.
- [5] Pollard, R. (2006). Home advantage in soccer: Variations in its magnitude and a literature review of the inter-related factors associated with its existence. *Journal of Sport Behavior*, 29(2), 169–189.